An-Najah National University

Course Assignment:Linux Commands & System Administration

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Introduction

System and Network Administration forms the backbone of modern computing infrastructures. The Linux Operating System serves as a vital platform for servers, network management, and enterprise administrative tools. Mastering the Linux Command Line Interface (CLI) provides system administrators with the necessary precision, speed, and efficiency to manage system hardware, oversee disk partitioning, handle user accounts, control file permission structures, and automate operational workflows.

This report demonstrates the practical application of fundamental Linux terminal commands extracted from the core modules of System Administration. It encompasses detailed execution logs, output simulations, and descriptions across key administrative domains, including:

1. **System Information & Hardware Management:** Auditing RAM usage, block devices, CPU specifications, and file system disk allocation.
2. **File and Directory Operations:** Managing the Linux file hierarchy, directory traversal, file copying, moving, and safe removal procedures.
3. **File Content Inspection & Text Processing:** Utilizing file viewers (`cat`, `less`), stream filters (`head`, `tail`), line counters (`wc`), and inline file editors (`vi`).
4. **Shell Environment & I/O Manipulation:** Working with shell execution history, environment variables, standard streams (`stdin`, `stdout`, `stderr`), file descriptors, and pipelines.

Part 1: System Information & Hardware Management

1. `free -m`

- **Description:** Displays the total, used, and free amount of physical memory (RAM) and swap memory in megabytes.
- **Command & Output:**

```
$ free -m
               total        used        free      shared  buff/cache   available
Mem:            7945        2104        3145         180        2695        5342
Swap:           2047           0        2047
```

2. `lsblk`

- **Description:** Lists information about all available block devices (hard drives, partitions) in a tree-like format.
- **Command & Output:**

```
$ lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
sda      8:0    0    50G  0 disk
├─sda1   8:1    0   500M  0 part /boot
└─sda2   8:2    0  49.5G  0 part /
sr0     11:0    1  1024M  0 rom
```

3. `lscpu`

- **Description:** Displays detailed information about the CPU architecture.
- **Command & Output:**

```
$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Byte Order:              Little Endian
CPU(s):                  2
On-line CPU(s) list:     0,1
Vendor ID:               GenuineIntel
CPU family:              6
Model name:              Intel(R) Core(TM) i5 CPU @ 2.67GHz
```

4. `df -h`

- **Description:** Shows the amount of disk space available on the file system in a human-readable format.
- **Command & Output:**

```
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        49G   14G   33G  30% /
/dev/sda1       487M  134M   324M  30% /boot
```

Part 2: File and Directory Operations

5. `ls -la`

- **Description:** Lists all files and directories in the current folder in a long format, including hidden files (starting with `.`).
- **Command & Output:**

```
$ ls -la
total 32
drwxr-xr-x. 4 user user 4096 Oct 13 10:00 .
drwxr-xr-x. 3 root root 4096 Oct 13 09:00 ..
-rw-r--r--. 1 user user  220 Oct 13 09:00 .bashrc
drwxr-xr-x. 2 user user 4096 Oct 13 09:30 Documents
```

6. `pwd`

- **Description:** Prints the absolute pathname of the current working directory.
- **Command & Output:**

```
$ pwd
/home/user
```

7. `mkdir dir1` & `cd dir1`

- **Description:** Creates a new directory named `dir1` and changes the current working directory to it.
- **Command & Output:**

```
$ mkdir dir1
$ cd dir1
$ pwd
/home/user/dir1
```

8. `cp file1.txt file2.txt`

- **Description:** Copies `file1.txt` and saves it as `file2.txt`.
- **Command & Output:**

```
$ echo "Hello Linux" > file1.txt
$ cp file1.txt file2.txt
$ cat file2.txt
Hello Linux
```

9. `mv file1.txt target_dir/`

- **Description:** Moves or renames files and directories.
- **Command & Output:**

```
$ mkdir target_dir
$ mv file1.txt target_dir/
$ ls target_dir/
file1.txt
```

10. `rm file1.txt / rmdir / rm -r`

- **Description:** Removes files or directories (including recursive deletion).
- **Command & Output:**

```
$ rm file2.txt
$ rmdir target_dir
rmdir: failed to remove 'target_dir': Directory not empty
$ rm -r target_dir
```

Part 3: Viewing and Filtering Files

11. `cat file1`

- **Description:** Displays the entire content of `file1` to the screen.
- **Command & Output:**

```
$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
```

12. `head -n 2 /etc/passwd`

- **Description:** Displays the first specified number of lines of a file.
- **Command & Output:**

```
$ head -n 2 /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
```

13. `tail -n 2 /etc/passwd`

- **Description:** Displays the last specified number of lines of a file.
- **Command & Output:**

```
$ tail -n 2 /etc/passwd
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
user:x:1000:1000:User:/home/user:/bin/bash
```

14. `wc /etc/passwd`

- **Description:** Counts the number of lines, words, and characters in a file.
- **Command & Output:**

```
$ wc /etc/passwd
  35  65 1500 /etc/passwd
```

15. `paste file1 file2`

- **Description:** Merges contents of two files line by line.
- **Command & Output:**

```
$ echo -e "A\nB" > f1.txt
$ echo -e "1\n2" > f2.txt
$ paste f1.txt f2.txt
A       1
B       2
```

Part 4: Shell Basics, History, and I/O Redirection

16. `history`

- **Description:** Views the list of previously entered commands.
- **Command & Output:**

```
$ history
  1  lsblk
  2  free -m
  3  pwd
  4  history
```

17. `printenv`

- **Description:** Displays all environmental variables.
- **Command & Output:**

```
$ printenv USER
user
```

18. `echo $PATH`

- **Description:** Displays the value of the PATH environment variable.
- **Command & Output:**

```
$ echo $PATH
/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin
```

19. `date > current_time.txt`

- **Description:** Redirects standard output to a file, overwriting its content.
- **Command & Output:**

```
$ date > current_time.txt
$ cat current_time.txt
Tue Oct 13 18:01:00 UTC 2026
```

20. `echo "Text" >> current_time.txt`

- **Description:** Appends standard output to the end of an existing file.
- **Command & Output:**

```
$ echo "System Check OK" >> current_time.txt
$ cat current_time.txt
Tue Oct 13 18:01:00 UTC 2026
System Check OK
```

21. `ps aux | grep bash`

- **Description:** Uses a pipeline to send the output of one command as input to another.
- **Command & Output:**

```
$ ps aux | grep bash
user      1234  0.0  0.1  11584  2850 pts/0    Ss   18:00   0:00 /bin/bash
```

Conclusion & Summary

This report summarizes the essential Linux commands and system administration concepts across four key areas:

1. **System & Hardware Audit:** Using commands like `free`, `lsblk`, `lscpu`, and `df` allows administrators to monitor hardware resources, memory, and disk space efficiently.
2. **File & Directory Management:** Command-line operations such as `ls`, `mkdir`, `cp`, `mv`, and `rm` enable fast navigation and full control over the Linux directory structure.
3. **File Viewing & Text Processing:** Utilities including `cat`, `head`, `tail`, `wc`, `paste`, and the `vi` editor provide powerful tools for reviewing logs and editing configuration files.
4. **Shell Administration & I/O Redirection:** Mastering environment variables (`printenv`), command history (`history`), standard streams, file descriptors, and pipelines (`|`) enables process automation and advanced shell scripting.

Overall, proficiency in these command-line tools is crucial for maintaining system stability, troubleshooting issues, and managing Linux infrastructure effectively.